

7-amaliy mashg'ulot. Rang va to'qima asosida tahlil

2026–2027 o'quv yili

Mavzu. Rang fazolari (RGB, HSV, CIE Lab), rang niqobi va k-means kvantlash (PSNR); to'qima belgilari: GLCM va Haralick belgilari, LBP, Gabor filtrlari banki, Laws energiya belgilari; to'qima bo'yicha tasniflash (SVM, k-NN) va segmentatsiya, precision, recall, F1-score.

Maqsadlar. Talaba: (1) HSV, ΔE_{76} , PSNR, GLCM, LBP, χ^2 ni qo'lda hisoblaydi; (2) GLCM, LBP va Laws belgilarining xossalari isbotlaydi; (3) `scikit-image` va `OpenCV` yordamida belgilarni hisoblaydi va qo'lda hisoblangan natija bilan solishtiradi; (4) to'qima tasniflagichini quradi va uni yoritish va burilishga chidamlilik bo'yicha halol baholaydi.

Baholash (100 ball). 1–16-topshiriqlar: har biri 2 ball (32); 17–30-topshiriqlar: har biri 4 ball (56); kod sifati, takrorlanuvchanlik (seed = 42, ikki marta ishga tushirishda bir xil natija) va qisqa xulosalar: 12 ball. Javoblar son qiymati bilan, kamida 4 xona aniqlikda (yoki topshiriqda ko'rsatilgan formatda) beriladi.

Muhit (setup)

Python 3.11 yoki 3.12, faqat CPU. Virtual muhit yaratib, qat'iy versiyalarni o'rnatib: `python -m venv .venv; pip install -r requirements.txt`, bunda `requirements.txt`: `numpy==2.4.4`, `scipy==1.17.1`, `pandas==3.0.2`, `scikit-learn==1.8.0`, `scikit-image==0.26.0`, `matplotlib==3.10.9`, `pillow==12.2.0`, `opencv-python-headless==4.13.0.92`, `torch==2.14.0`, `torchvision==0.29.0`. Barcha ma'lumotlar offlayn: `skimage.data` (brick, grass, gravel, astronaut) va sintetik tasvirlar. Barcha kodni bitta `solutions.py` faylida (har topshiriqqa bitta funksiya va `main`) yozing; fayllarni `./out/` ga saqlang; `python -W error::DeprecationWarning solutions.py` xatosiz ishlashi shart.

Umumiy yordamchi kod (25–26, 28–29-topshiriqlarda ishlatiladi):

```
import numpy as np, cv2
from skimage import data
from skimage.feature import graycomatrix, graycoprops, local_binary_pattern
SEED = 42
def texture_patches(size=64):
    """brick/grass/gravel (512x512) -> 64x64 yamoqlar (har sinfda 64 ta).
    Yuqori yarmi (r < 256) -- train (96 ta), pastki yarmi -- test (96 ta)."""
    X, y, is_tr, pos = [], [], [], []
    for c, name in enumerate(("brick", "grass", "gravel")):
        I = getattr(data, name)() # uint8, 512x512
        for r in range(0, 512, size):
            for q in range(0, 512, size):
                X.append(I[r:r+size, q:q+size]); y.append(c)
                is_tr.append(r < 256); pos.append((c, r, q))
    return np.array(X), np.array(y), np.array(is_tr), pos
def lbp_hist(p, P=8, R=1): # riuz ('uniform') gistogramma, P+2 katak
    L = local_binary_pattern(p, P, R, method="uniform").astype(int)
    h = np.bincount(L.ravel(), minlength=P + 2); return h / h.sum()
```

1. Qo'lda hisoblash, hosil qilish va tushunchalar

Belgilashlar: $A = \begin{pmatrix} 0 & 1 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 0 & 2 & 3 & 3 \end{pmatrix}$ ($L = 4$ daraja) — 6–8-topshiriqlar uchun; $\delta = (\Delta r, \Delta c)$ siljish, $(0, 1)$ — “o'ngga 1 piksel”, $(1, 0)$ — “pastga 1 piksel”. LBP: kvadrat 8-qo'shnilik, $p = 0$ — o'ng qo'shni, so'ng soat strelkasiga teskari; $s(x) = 1$ ($x \geq 0$). Entropiya — bitlarda ($\log_2$).

- HSV.** $(R, G, B) = (200, 150, 50)$ uchun H (gradus), S , V ni hisoblang va `OpenCV` `uint8` shkalasidagi (H_{cv}, S_{cv}, V_{cv}) ni yozing.
- Yoritish.** a) Barcha kanallar $\alpha > 0$ ga ko'paytirilganda (kesilishsiz) H va S o'zgarmasligini isbotlang; $(100, 75, 25)$ misolida tekshiring. b) Sarg'ish yoritishda kanallar $(1, 0; 0, 8; 0, 6)$ ga ko'paytiriladi: $(200, 150, 50)$ uchun yangi H , S va `OpenCV` qiymatlari.
- Lab.** $(L^*, a^*, b^*)_1 = (50, 20, -10)$, $(L^*, a^*, b^*)_2 = (55, 14, -2)$: ΔE_{76} ni toping. Farq “zo'rg'a sezilarli” ($\approx 2,3$) chegaradan necha barobar katta? Qaysi koordinata eng katta hissa qo'shadi?
- PSNR.** a) $MSE = 50$ da PSNR; b) PSNR = 40 dB da MSE; c) MSE ikki barobar kamaysa PSNR necha dB ortadi? (8 bitli tasvir.)

5. **Lloyd (k-means), 1D.** Yorqinliklar $\{2, 4, 10, 12, 30\}$, $k = 2$, boshlangʻich markazlar $(2, 4)$. Har bir iteratsiyadagi tayinlash va markazlarni yozing, toʻxtash momentini, J , MSE va PSNR ni toping. Bu yechim global minimummi? (1D da faqat ketma-ket boʻlinishlarni tekshirish yetarli.)
6. **GLCM.** A tasvir, $\delta = (0, 1)$: C , simmetrik normallangan $P = (C + C^T) / \sum$ va contrast, homogeneity, ASM, energy, entropy (bit) ni hisoblang.
7. **Yoʻnalish.** A uchun $\delta = (1, 0)$ boʻyicha nosimmetrik C ni tuzing. Normallangan nosimmetrik matritsalar boʻyicha $(0, 1)$ va $(1, 0)$ uchun contrast ni solishtiring: qaysi yoʻnalishda tasvir “silliqlik”?
8. **Hosil qilish.** Simmetrik normallangan GLCM uchun contrast $= 2\sigma^2(1 - \text{correlation})$ ekanini isbotlang ($\mu_i = \mu_j = \mu$, $\sigma_i = \sigma_j = \sigma$). 6-topshiriq uchun μ , σ^2 , correlation ni hisoblab, tenglikni tekshiring.
9. **Chegaralar.** a) Oʻzgarasiz tasvir uchun contrast, homogeneity, ASM, entropy qanday? Nega correlation aniqlanmagan (va **graycoprops** nima qaytaradi)? b) P ning nolmas elementlari soni n boʻlsa, $1/n \leq \text{ASM} \leq 1$ va $0 \leq \text{entropy} \leq \log_2 n$ ni isbotlang.
10. **LBP.** 3×3 yamoq $\begin{pmatrix} 8 & 9 & 7 \\ 6 & 6 & 5 \\ 2 & 1 & 4 \end{pmatrix}$ ($g_c = 6$) uchun $s_0, \dots, s_7$, LBP, oʻtishlar soni U , LBP^{riu2} va minimal aylanma kodni toping. Chap qoʻshni 6 shovqin tufayli 5 ga aylansa, LBP va riu2 qanday boʻladi?
11. **LBP invariantligi (tushuncha).** `local_binary_pattern(I, 8, 1, 'uniform')` (diagonal qoʻshnilar bilinear interpolatsiya qilinadi) uchun quyidagi almashtirishlarning qaysilarida kodlar *aynan* saqlanadi: (a) $2I + 10$; (b) $1000\sqrt{I}$; (c) $255 - I$; (d) 90° ga burish? Har birini asoslang ((b) ga alohida eʼtibor bering).
12. **Uniform naqshlar soni.** P bitli aylanma satrda $U \leq 2$ boʻlgan naqshlar soni $P(P-1)+2$ ekanini isbotlang. $P = 8$ va $P = 16$ uchun **uniform** (riu2) va **nri_uniform** gistogrammalarining kataklari sonini toping.
13. **χ^2 masofa.** $\mathbf{a} = (0,5; 0,25; 0,25)$, $\mathbf{b} = (0,25; 0,25; 0,5)$: $\chi^2(\mathbf{a}, \mathbf{b}) = \sum_k (a_k - b_k)^2 / (a_k + b_k)$ va $\|\mathbf{a} - \mathbf{b}\|_2$ ni hisoblang.
14. **Gabor.** a) Bir oktavali filtr uchun $\sigma/\lambda = \frac{1}{\pi}\sqrt{\ln 2/2} \cdot 3$; $\lambda = 16$ da σ ni toping. b) $I = \cos(2\pi y/8)$ (gorizontal yoʻllar) uchun qaysi (θ, λ) filtr eng katta javob beradi? c) Nega Gabor energiyasi $E = (I * g_{\cos})^2 + (I * g_{\sin})^2$ naqsh fazasiga (yoʻllarning siljishiga) deyarli bogʻliq emas?
15. **Laws.** $\text{L5} = (1, 4, 6, 4, 1)$, $\text{E5} = (-1, -2, 0, 2, 1)$. $\text{E5L5} = \text{E5}^T \text{L5}$ niqobining birinchi va uchinchi qatorini yozing, elementlar yigʻindisini toping. 5×5 “rampa” tasvirlar $I(r, c) = r$ va $I(r, c) = c$ ($r, c = 0..4$) uchun markazdagi javobni (korrelyatsiya, yaʼni $\sum M \odot I$) hisoblang. Niqob qaysi chegaralarni aniqlaydi?
16. **Baholash.** Qator — haqiqiy sinf (brick, grass, gravel), ustun — bashorat: $C = \begin{pmatrix} 28 & 3 & 1 \\ 2 & 24 & 6 \\ 0 & 4 & 28 \end{pmatrix}$. Accuracy, har bir sinf uchun precision, recall, F1-score va macro-F1 ni toping.

2. Dasturlash topshiriqlari

Har bir topshiriq — `solutions.py` dagi alohida funksiya (`task17_...()`, ..., `task30_...()`). Sonlarni koʻrsatilgan formatda chop eting.

17. **RGB→HSV.** `rgb_to_hsv_manual(rgb) -> ndarray: uint8 (... , 3)` massivdan (H°, S, V) (`float64`, $S, V \in [0, 1]$) ni vektorlashtirib hisoblang ($\Delta = 0$ da $H = 0$). Sinov: `rng = np.random.default_rng(42); rgb = rng.integers(0, 256, (1, 1000, 3), dtype=np.uint8); etalon: cv2.cvtColor(rgb.astype(np.float32)/255, cv2.COLOR_RGB2HSV) (float32 kirishda H gradusda). Chiqish: max|dH| (aylanma farq), max|dS|, max|dV| — %.2e formatda.`
18. **Rang niqobi.** 120×120 kulrang (128) fon, markazi (60,60), radiusi 30 boʻlgan toʻq sariq (255,140,0) doira (etalon niqob). Kechki tasvir: kanallar (0,8; 0,6; 0,4) ga koʻpaytirilib, yaxlitlanadi. RGB niqobi: `cv2.inRange(x, (225,110,0), (255,170,30))`; HSV niqobi: `cv2.inRange(hsv, (5,150,40), (25,255,255))`. Kunduzi va kechqurun uchun IoU (%.4f) va obyekt/fon HSV qiymatlarini chop eting. Nega S ning quyi chegarasi 150 olingan?
19. **ΔE_{76} .** `skimage.color.rgb2lab` va `deltaE_cie76` bilan juftliklar: (60,180,75)–(51,117,34); (200,150,50)–(100,75,25); (128,128,128)–(130,128,128); (255,0,0)–(0,0,255). Lab (2 xona) va ΔE_{76} (%.2f). 2-topshiriqdagi HSV xulosasi bilan solishtiring.
20. **k-means kvantlash.** `img = data.astronaut()[::4, ::4]` ($128 \times 128 \times 3$). `quantize(img, k): KMeans(n_clusters=k, n_init=4, random_state=42)`, palitra `np.round` bilan `uint8` ga keltiriladi. $k \in \{2, 4, 8, 16, 32\}$ uchun MSE va PSNR (%.2f) jadvali; $k = 8$ natijasini `out/m07_kmeans_k8.png` ga saqlang. k ikki barobar oshganda PSNR oʻsishi qanday oʻzgaradi?
21. **GLCM qoʻlda.** `glcm_manual(I, dr, dc, L, symmetric=False, normed=False)` ni faqat NumPy bilan yozing. A uchun (0,1), (1,0), (1,1) siljishlarda `graycomatrix(A, [1], [angle], levels=4)` bilan tengligini (True/False) tekshiring: har bir siljishga mos `angle` ni oʻzingiz toping. 6-topshiriq belgilarini chop eting; `graycoprops(..., 'entropy')` bilan farqni tushuntiring.
22. **GLCM jadvali.** `glcm_vector(p): q = p // 8 (L = 32)`, `graycomatrix(q, [1, 3], [0, pi/4, pi/2, 3pi/4], levels=32, symmetric=True, normed=True)`; contrast, homogeneity, energy, correlation ning 4 yoʻnalish boʻyicha oʻrtachasi (8 oʻlchov). `brick, grass, gravel` (butun tasvir) uchun $d = 1, 3$ jadval chiqaring.
23. **LBP funksiyasi.** `lbp_square(patch) -> (s, code, U, riu2, min_ror)` (10-topshiriq qoidasi). Sinov: 10-

topshiriq yamog'i, $3g + 7$, chap qo'shni 5 bo'lgan variant, o'zgarmas yamoq.

24. **LBP invariantligi.** `I = data.brick()`, `local_binary_pattern(J, 8, 1, method='uniform')`; ichki piksellar `[[1:-1, 1:-1]]` bo'yicha bir xil kodlar ulushi (%.4f): `2*I.astype(np.uint16)+10; np.round(1000*np.sqrt(I)).astype(np.uint16)`; `255-I`; `np.rot90(I)` (kod xaritasini qaytarib burang). Butun sonli tiplardan foydalaning (`float` kirish ogohlantirish beradi). `brick` ning 10 katakli `riu2` gistogrammasini chiqaring. 11-topshiriq javobi bilan solishtiring.
25. **To'qima tasnifi (SVM).** `texture_patches()`; belgilar: (i) o'rtacha va std; (ii) `lbp_hist(p) + lbp_hist(p, 16, 2)` (28); (iii) `glcm_vector(8)`; (iv) (ii)+(iii). Tasniflagich: `make_pipeline(StandardScaler(), SVC(kernel='rbf', C=10, gamma='scale'))`. Har biri uchun test accuracy ($k/96$) va macro-F1, (iv) uchun chalkashlik matritsasi. Nega train/test tasvirning turli yarmidan olinadi?
26. **Chidamlilik.** 25-topshiriqdagi modellarni o'qiting (train o'zgarmaydi), lekin test yamoqlarini uch variantda sinang: asl; `np.round(0.6*p+40).astype(np.uint8)`; `np.rot90(p)`. Belgilar: o'rtacha+std; LBP (28); faqat $\theta = 0$ bo'yicha GLCM (`angles=[0]`, $d = 1, 3, 8$ o'lchov); `glcm_vector`. 4×3 accuracy jadvali. Qo'shimcha: butun `brick` uchun $d = 1$ contrast ($\theta = 0$ va 4 yo'nalish o'rtachasi) burishdan oldin va keyin.
27. **Gabor bilan yo'nalish.** 128×128 tasvir $I = \cos(2\pi(x \cos 60^\circ + y \sin 60^\circ)/10)$ (`np.mgrid`). Bank: $\theta = 0^\circ, 15^\circ, \dots, 165^\circ$, `cv2.getGaborKernel((31, 31), 5.6, theta, 10, 0.5, psi)`, $\psi \in \{0, \pi/2\}$, `cv2.filter2D(img, cv2.CV_64F, k)`; energiya — $rc^2 + rs^2$ ning `[32:-32, 32:-32]` bo'yicha o'rtachasi, maksimumga normallangan (%.3f). Dominant θ ni toping.
28. **Gabor + k-NN.** Har bir to'qima tasvirida (butun 512×512) $\lambda \in \{4, 8, 16\}$, $\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$, $\sigma = 0,56\lambda$, $\gamma = 0,5$, `ksize = 2[3σ] + 1`, `borderType=cv2.BORDER_REFLECT` bilan amplituda $\sqrt{rc^2 + rs^2}$ xaritalarini hisoblang, so'ng har bir yamoq bo'yicha o'rtacha va std (24 belgi). `StandardScaler` + `KNeighborsClassifier(n_neighbors=k)`, $k \in \{1, 3, 5\}$: accuracy va macro-F1.
29. **Laws + LinearSVC.** Tasvirdan lokal o'rtachani ayiring (`scipy.ndimage.uniform_filter(I, 15, mode='reflect')`), 16 ta 5×5 niqob bilan `cv2.filter2D(BORDER_REFLECT)`, modul oling; simmetrik juftlarni o'rtachalab 9 ta xarita: E5E5, S5S5, R5R5, E5L5+L5E5, S5L5+L5S5, R5L5+L5R5, S5E5+E5S5, R5E5+E5R5, R5S5+S5R5. Belgi — yamoq bo'yicha o'rtacha. `StandardScaler` + `LinearSVC(C=1.0, max_iter=20000, random_state=42)`: har sinf uchun o'rtacha belgi (train), accuracy, chalkashlik matritsasi.
30. **To'qima segmentatsiyasi.** Mozaika: `base = data.brick()[ :256, :256]`, markazi (128,128), radiusi 80 bo'lgan doirada `data.grass()[ :256, :256]` piksellari. LBP (8,1, uniform) xaritasi; 16×16 bloklar (256 ta); har bir blok uchun markazi blok markazida bo'lgan $W \times W$ oynadagi 10 katakli gistogramma (LBP xaritasi `np.pad(..., (W-16)//2, mode='reflect')`), $W \in \{16, 32\}$. `KMeans(n_clusters=2, n_init=10, random_state=42)`; blok etaloni — doira ulushi $> 0,5$. Blok accuracy (yorliq almashishini hisobga oling) va `out/m07_segment.png`. Nega katta oyna yaxshiroq?

Hujjatlar

<https://scikit-image.org/docs/stable/api/skimage.feature.html> (graycomatrix, graycoprops, local_binary_pattern); <https://scikit-image.org/docs/stable/api/skimage.color.html> (rgb2lab, deltaE_cie76); <https://scikit-image.org/docs/stable/api/skimage.data.html>; https://docs.opencv.org/4.x/de/d25/imgproc_color_conversions.html; https://docs.opencv.org/4.x/d4/d86/group_imgproc_filter.html (getGaborKernel, filter2D); https://docs.opencv.org/4.x/d2/de8/group_core_array.html (inRange); <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.svm.SVC.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.svm.LinearSVC.html>; <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>; https://scikit-learn.org/stable/modules/model_evaluation.html; https://docs.scipy.org/doc/scipy/reference/generated/scipy.ndimage.uniform_filter.html.